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BCMCMCS 201

**Second Semester B.Com. Degree Examination, June/July 2025
(SEP) (2024 – 25 Batch Onwards)
QUANTITATIVE TECHNIQUES – II**

Time : 3 Hours

Max. Marks : 80

SECTION – AAnswer **any four** questions.**(4×4=16)**

1. Define correlation. Explain different types of correlation.
2. For bivariate data, the two regression coefficients are -7.3 and -0.11 . Find the correlation coefficient.
3. What is time series ? Explain the utility of time series analysis.
4. A construction company is bidding for two contracts A and B. The probability that it will get contract A is $3/5$ and B is $1/4$ and the probability that it getting both is $1/8$. What is the probability that it will get contract A or B ?
5. A ratio in its simplest form is $5 : 7$. If the difference between the quantities is 12, find the quantities.
6. In how many ways can letter "ACCOUNTS" be arranged ?

SECTION – BAnswer **any four** questions.**(4×8=32)**

7. Find the rank correlation coefficient between X and Y.

X	40	15	30	40	40	20	30
Y	50	100	50	45	40	70	48

P.T.O.



8. Following is the distribution of students according to their height and weight :

Height	57	59	62	63	64	65	55	58	57	60
Weight	113	117	126	126	130	129	111	116	112	120

Obtain correlation coefficient.

9. Compute 2 yearly centred moving average for the following data.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024
Demand	208	212	210	215	213	220	224	210	214

10. A bag contains 6 red and 4 white balls. Two balls are drawn from the bag. What is the probability that they are of the same colour ?
11. If 40 farmers working 10 hours daily can harvest a field in 15 days, how long it will take for 30 farmers working 12 hours daily to do the same work ?
12. Among 8 men and 6 women in a group, 4 has to be selected for an event such that in selection there should be 2 women. In how many ways the team can be selected ?

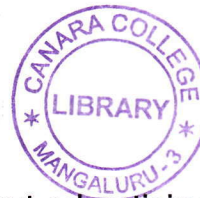
SECTION – C

Answer **any two** questions.

(2×16=32)

13. Compute the coefficient of correlation between dividends and price of securities given below.

Security Price	Annual Dividend				
	0-5	5-10	10-15	15-20	20-25
50-100	5	6	-	-	-
100-150	-	5	8	1	-
150-200	-	-	4	3	3
200-250	-	-	-	6	3
250-300	-	-	-	4	2



14. You are given the following information about advertising and sales :

	Advertising Expenses (X) (in Rs. Crores)	Sales (Y) (in Rs. Crores)
Mean	10	90
Standard Deviation	3	12
Coefficient of correlation = 0.8		

- Obtain two regression equations.
- What should be the advertisement expenditure if the company wants to attain sales target of Rs. 150 crores ?
- Find the likely sales when advertisement expenditure is Rs. 50 crores.

15. Compute trend values by the method of least squares from the following data. Also estimate the sales for the year 2026.

Year	2017	2018	2019	2020	2021	2022	2023	2024
Sales	104	108	110	106	108	101	112	115

16. a) Solve :

$$3x + 5y = 84$$

$$5x + 4y = 88$$

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b) One card is drawn from a well shuffled pack of (52) cards. What is the probability that the card will be

i) A Red or a King

ii) A King or a Spade

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